

Problem 1

KK 3.19 - Series and Parallel Springs

We are asked to find the oscillation frequency for (a) series and (b) parallel springs with spring constants k_1 and k_2 .

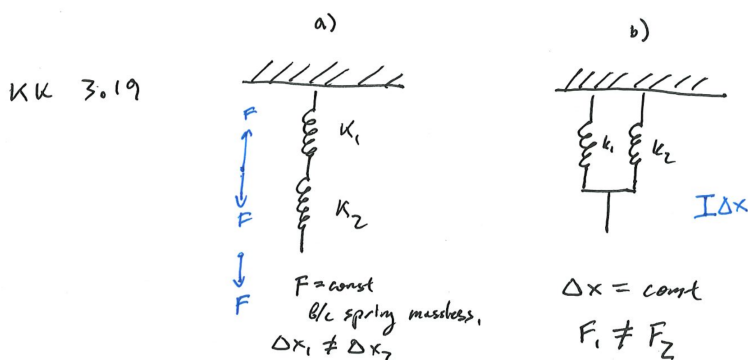
First let us recall

$$m\ddot{x} = F = -kx \quad (1.1)$$

is an equation of the form

$$\ddot{x} + \omega^2 x = 0 \quad (1.2)$$

with $\omega = \sqrt{k/m}$ as the oscillation frequency.



simplified case: identical springs

Our task is to find $k_{effective}$ as a function of k_1 and k_2 . When approaching an unknown problem, it is helpful to simplify the problem. What happens if $k_1 = k_2$? Then the springs are identical, and they must have identical displacements.

So in the parallel case, the sum of forces is

$$m\ddot{x} = -2kx \quad (1.3)$$

suggesting the parallel frequency is $\omega_p = \sqrt{2k/m}$. In the series case, the overall displacement split across two identical springs must be identical. So the individual displacement is $\Delta x/2$. This suggests $k_{eff,s} = k/2$ such that the series frequency is $\omega_s = \sqrt{k/2m}$. At this point one might recognize that series/parallel springs appear to behave like series/parallel capacitors and guess the form

$$k_p = k_1 + k_2 \quad (1.4)$$

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} \quad (1.5)$$

This makes intuitive sense that stretching two springs requires twice the force as stretching one, and stretching a spring that is doubled in length requires half the relative displacement ($\Delta x/L_s$), resulting in half the force.

full problem

Now let's approach the full problem.

For the parallel case, the displacement of the two springs is identical. Therefore Newton's law gives

$$m\ddot{x} = -k_1x - k_2x = -(k_1 + k_2)x \quad (1.6)$$

We find

$$\ddot{x} + \frac{k_1 + k_2}{m}x = 0 \quad (1.7)$$

which gives

$$\boxed{\omega_p = \sqrt{\frac{k_1 + k_2}{m}}} \quad (1.8)$$

For the series case, the two springs must have the same force

$$F = k_1x_1 = k_2x_2 \quad (1.9)$$

such that $x_1 = F/k_1$ and $x_2 = F/k_2$. But the total displacement is $x_T = x_1 + x_2$. Therefore

$$x_T = x_1 + x_2 = F \left(\frac{1}{k_1} + \frac{1}{k_2} \right) \quad (1.10)$$

So we find

$$F = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}} x_T \quad (1.11)$$

which satisfies the form

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} \quad (1.12)$$

with $F = k_s x_T$. Therefore

$$\boxed{\omega_s = \sqrt{\frac{k_1 k_2}{m(k_1 + k_2)}}} \quad (1.13)$$

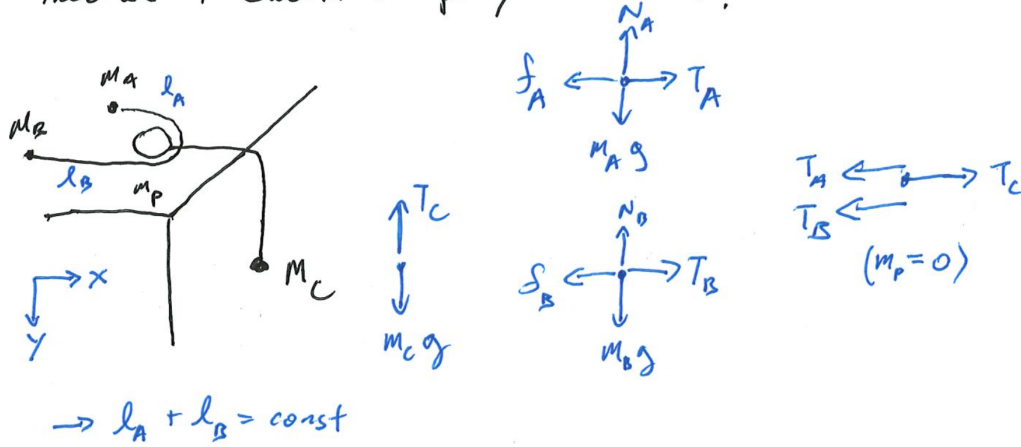
To summarize, parallel springs have equal displacement but different forces, while series springs have equal forces and different displacement.

KK 3.7 - Pulley and Rope with Friction

This problem builds on KK 2.10.

KK 3.7

There are 4 blocks. The pulley is massless.



simplified case: identical upper blocks

Again, we may start by solving a simpler problem. What if $m_A = m_B$? Then the pulley acts on both equally, and the two blocks move as a unit. We see

$$\Delta x_A = \Delta x_B = \Delta x_C \quad (1.14)$$

All three blocks have equal linear displacement. Then the resistive force is Friction

$$f = \mu N = \mu(m_A + m_B)g \quad (1.15)$$

The accelerations are related through the displacement

$$\ddot{x}_A = \ddot{x}_B = \ddot{y}_C \quad (1.16)$$

and observing the lower block we find

$$\begin{aligned} m_C \ddot{y}_C &= \sum F_y = m_C g - T \\ &= m_C g - \mu(m_A + m_B)g \end{aligned} \quad (1.17)$$

where tension T is equal to the friction from the upper stage. Now at the upper blocks

$$(m_A + m_B) \ddot{x}_A = \sum F_x = T - \mu(m_A + m_B)g \quad (1.18)$$

We can use the 3 constraints (Eqs 1.16 - 1.18) to eliminate acceleration and solve for tension

$$\ddot{y}_C = g - \frac{T}{m_C} \quad (1.19)$$

$$\ddot{x}_A = \frac{T}{m_A + m_B} - \mu g \quad (1.20)$$

Setting these equal to each other shows

$$(1 + \mu)g = T \left(\frac{1}{m_C} + \frac{1}{m_A + m_B} \right) \quad (1.21)$$

Therefore

$$T = \frac{g(1 + \mu)}{\frac{1}{m_C} + \frac{1}{m_A + m_B}} \quad (1.22)$$

which can be rearranged into the more dimensionless form

$$T = \left(\frac{1 + \mu}{1 + \frac{m_C}{m_A + m_B}} \right) m_C g \quad (1.23)$$

full case

The difference when $m_A \neq m_B$ is that the two blocks may move separately under the action of the pulley. We can consider another simplifying limit, where $m_A \rightarrow \infty$. Then m_A is stuck to the platform, and only Δx_B moves in response to Δy_C . But the opposite must be true if $m_B \rightarrow \infty$. This suggests a symmetric form

$$\Delta x_A + \Delta x_B = 2\Delta y_C \quad (1.24)$$

We see in the limit where one mass is fixed, that the other mass must move twice this distance. Indeed, this is constrained by the length of the pulley string.

Another way to reach this conclusion is to consider the motion of the pulley, $\Delta x_p = \Delta y_C$. It follows that the motion of the blocks relative to the pulley must be matched

$$(\Delta x_A - \Delta x_p) + (\Delta x_B - \Delta x_p) = 0 \quad (1.25)$$

which also yields Eq 1.24. Therefore the accelerations are related by

$$\boxed{\ddot{x}_A + \ddot{x}_B = 2\ddot{y}_C} \quad (1.26)$$

For the tension in the rope we apply Newton's law to the three masses individually

$$m_A \ddot{x}_A = \sum F_x^A = T_A - \mu m_A g \quad (1.27)$$

$$m_B \ddot{x}_B = \sum F_x^B = T_B - \mu m_B g \quad (1.28)$$

$$m_C \ddot{y}_C = \sum F_y^C = m_C g - T_C \quad (1.29)$$

Here we use $\sum F_x^A$ to mean the sum of x-directed forces on mass A. Dividing out the masses gives

$$\ddot{x}_A = \frac{T_A}{m_A} - \mu g \quad (1.30)$$

$$\ddot{x}_B = \frac{T_B}{m_B} - \mu g \quad (1.31)$$

$$\ddot{y}_C = g - \frac{T_C}{m_C} \quad (1.32)$$

We can recognize that the pulley is massless to constrain

$$m_p \ddot{x}_p = 0 = T_C - (T_A + T_B) \quad (1.33)$$

But we can also use the fact that the pulley string is massless to constrain $T_A = T_B$. Therefore each upper block feels half the tension T of the lower block. Summing Eqs 1.30 and 1.31 gives

$$\ddot{x}_A + \ddot{x}_B = \frac{T_C}{2} \left(\frac{1}{m_A} + \frac{1}{m_B} \right) - 2\mu g = 2\ddot{y}_C \quad (1.34)$$

Therefore

$$\ddot{y}_C = \frac{T_C}{4} \left(\frac{1}{m_A} + \frac{1}{m_B} \right) - \mu g \quad (1.35)$$

We may combine this with Eq 1.32 to find

$$(1 + \mu)g = T_C \left[\frac{1}{m_C} + \frac{1}{4} \left(\frac{1}{m_A} + \frac{1}{m_B} \right) \right] \quad (1.36)$$

Therefore

$$T_C = \frac{(1 + \mu)g}{\frac{1}{m_C} + \frac{1}{4} \left(\frac{1}{m_A} + \frac{1}{m_B} \right)} \quad (1.37)$$

Let us recast the solution as

$$T_C = \left[\frac{1 + \mu}{1 + \frac{1}{4} \left(\frac{m_C}{m_A} + \frac{m_C}{m_B} \right)} \right] m_C g \quad (1.38)$$

which more neatly expresses the dimensionless relations involved. It is straight forward to confirm that this reduces to the simple case ($m_A = m_B$, Eq 1.23)

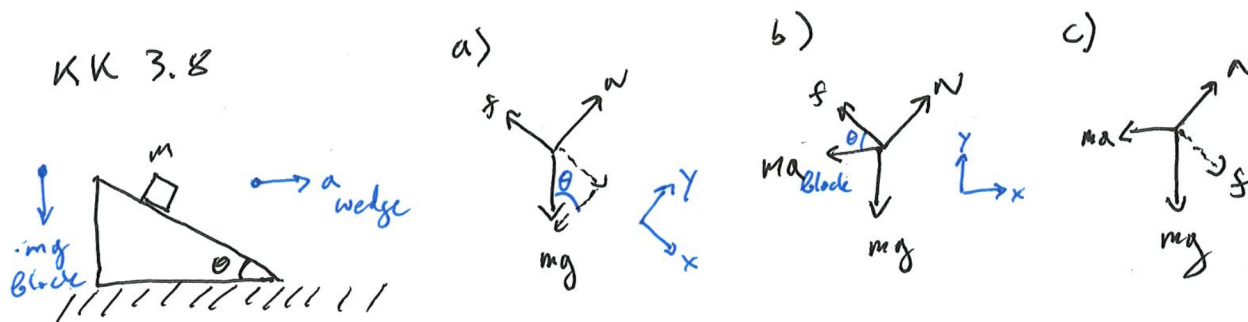
$$\frac{1}{4} \left(\frac{1}{m_A} + \frac{1}{m_B} \right) = \frac{1}{2m_A} = \frac{1}{m_A + m_B} \quad (1.39)$$

Also, in the friction-free case ($\mu \rightarrow 0$) we recover the solution from KK 2.10.

Problem 2

KK 3.8 - Block on Wedge with Friction

This problem builds on KK 2.9.



A block rests on an inclined wedge. In this problem the wedge is fixed ($M \rightarrow \infty$), else we would need to consider conservation of momentum in the x-direction.

a) friction and gravity

We are asked to find the coefficient of friction that allows the block to set at rest for a given wedge angle $\mu(\theta)$.

Let us adopt coordinates where $\hat{x}$ points down parallel to the slope, and $\hat{y}$ is up normal to the slope. Then

$$m\ddot{x} = \sum F_x = mg\sin\theta - f \quad (2.40)$$

and

$$m\ddot{y} = 0 = N - mg\cos\theta \quad (2.41)$$

Using the second constraint we find $f = \mu N = \mu mg\cos\theta$. Then the acceleration is

$$\begin{aligned} \ddot{x} &= g\sin\theta - \frac{f}{m} \\ &= g\sin\theta - g\mu\cos\theta \\ &= g\sin\theta \left[1 - \frac{\mu}{\tan\theta} \right] \end{aligned} \quad (2.42)$$

For the block to sit at rest, we need $\ddot{x} = 0$, or

$$\boxed{\mu > \tan\theta} \quad (2.43)$$

b) friction and minimum acceleration

Now the wedge is pushed to the right with acceleration a . This creates an equal and opposite force that tends to push the block left and up the ramp. If $\mu < \tan\theta$, then friction alone is not sufficient to stop the block from falling due to gravity. But the additional push from a can compensate. We are asked to find the minimum a .

We may solve this problem in the tilted (x,y) coordinates as before, but it is simpler to go to horizontal coordinates parallel to the ground.

In the x-direction there is finite acceleration

$$ma = \sum F_x = N\sin\theta - f\cos\theta \quad (2.44)$$

In the y-direction we demand zero acceleration

$$0 = \sum F_y = N\cos\theta + f\sin\theta - mg \quad (2.45)$$

Letting $f = \mu N$ we find

$$N = \frac{mg}{\cos\theta + \mu\sin\theta} \quad (2.46)$$

Therefore

$$\begin{aligned} a &= \frac{N}{m}(\sin\theta - \mu\cos\theta) \\ &= g \left(\frac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta} \right) \end{aligned} \quad (2.47)$$

Rearranging terms we conclude

$$\boxed{a \geq g \left(\frac{\tan\theta - \mu}{1 + \mu\tan\theta} \right)} \quad (2.48)$$

c) friction and maximum acceleration

Next we're asked to find the maximum acceleration before friction is overwhelmed.

Conceptually this is just a map ($\mu \mapsto -\mu$), because in (b) friction points up the slope to counteract gravity, and now friction points down the slope to counteract acceleration.

We write down right away

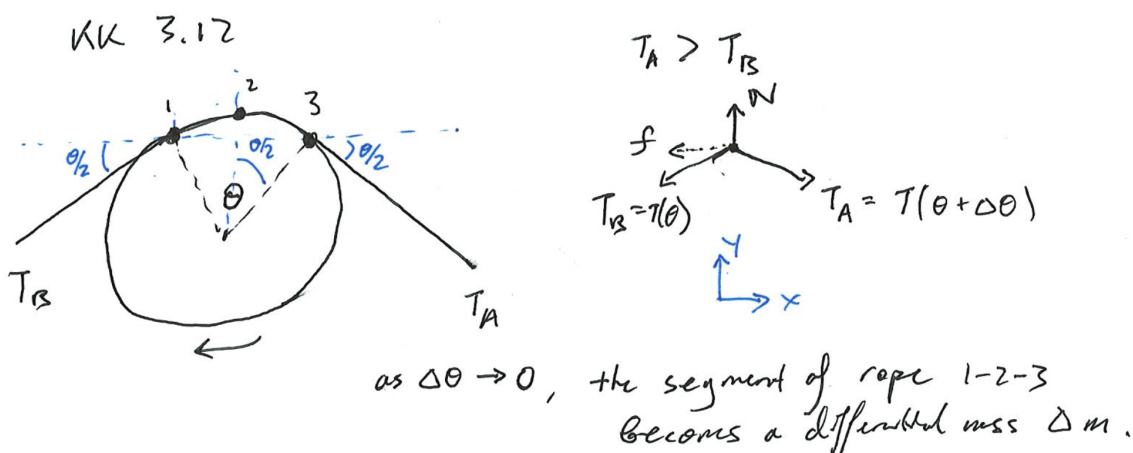
$$a \leq g \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right) \quad (2.49)$$

Problem 3

KK 3.12 - Capstan

We are asked to explain why a capstan, under the combined action of friction and tension, provides force multiplication that is exponential in angle

$$T_B = T_A e^{-\mu \theta} \quad (3.1)$$



differential form

This expression implies the differential relation

$$\frac{dT}{d\theta} = -\mu T \quad (3.2)$$

Let us consider the force acting on a small section of rope.

$$0 = \sum F_x = T_2 \cos \frac{\Delta \theta}{2} - T_1 \cos \frac{\Delta \theta}{2} - f \quad (3.3)$$

and

$$0 = \sum F_y = N - (T_1 + T_2) \sin \frac{\Delta \theta}{2} \quad (3.4)$$

Both forces sum to zero because the rope is stationary. We defined $T_2 > T_1$ such that $f = \mu N$ compensates for the difference in tension. Now letting $T_1 = T(\theta)$ and $T_2 = T(\theta + \Delta\theta)$ we find in the limit $\Delta\theta \rightarrow 0$ that

$$\lim_{\Delta\theta \rightarrow 0} \frac{T(\theta + \Delta\theta) - T(\theta)}{\Delta\theta} = \frac{dT}{d\theta} \quad (3.5)$$

and

$$\lim_{\Delta\theta \rightarrow 0} T(\theta + \Delta\theta) + T(\theta) = 2T(\theta) \quad (3.6)$$

Then under the small angle limit $\sin x \rightarrow x$

$$N = 2T \sin \frac{\Delta\theta}{2} = T\Delta\theta \quad (3.7)$$

While using $\cos x \rightarrow 1$ the other balance yields

$$\begin{aligned} T_2 - T_1 &= \frac{f}{\cos(\Delta\theta/2)} \\ T(\theta + \Delta\theta) - T(\theta) &= \mu N \\ &= \mu T \Delta\theta \end{aligned} \quad (3.8)$$

We divide both sides by $\Delta\theta$, take the limit $\Delta\theta \rightarrow 0$ and conclude

$$\frac{dT}{d\theta} = \mu T \quad (3.9)$$

integration

This expression can be integrated

$$\int \frac{dT}{T} = \mu \int d\theta \quad (3.10)$$

to find

$$\ln \left(\frac{T_A}{T_B} \right) = \mu(\theta_A - \theta_B) \quad (3.11)$$

Recall that the problem statement defines $T_A > T_B$. Therefore

$$T_B = T_A e^{-\mu(\theta_A - \theta_B)} \quad (3.12)$$

The capstan itself does not care whether $T_B > T_A$. If the winds on your sailboat changed such that the other side had greater tension, the friction would simply change directions, and the opposite direction would also hold an exponential tension.

Problem 4

KK 4.1 - Center of Mass of non-uniform rod

We're given a linear mass density

$$\lambda(x) = A \cos \left(\frac{\pi x}{2\ell} \right) \quad (4.1)$$

It's clear that A has dimensions of mass per length.

a) total mass

The total mass is

$$\begin{aligned}
 M &= \int \lambda dx = A \int_0^\ell \cos\left(\frac{\pi x}{2\ell}\right) dx \\
 &= A \frac{2\ell}{\pi} \left[\sin\left(\frac{\pi x}{2\ell}\right) \right]_0^\ell \\
 &= \frac{2}{\pi} A \ell
 \end{aligned} \tag{4.2}$$

b) center of mass

The center of mass is

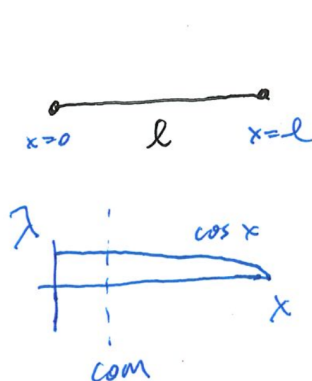
$$X = \frac{\int x \lambda dx}{\int \lambda dx} = \frac{A}{M} \int_0^\ell x \cos\left(\frac{\pi x}{2\ell}\right) dx \tag{4.3}$$

This integral can be solved with Integration by Parts. Recall $(uv)' = u'v + uv'$, then $\int uv' = uv - \int u'v$. We let $u = x$ and $v = \cos$. Then $u' = 1$ and

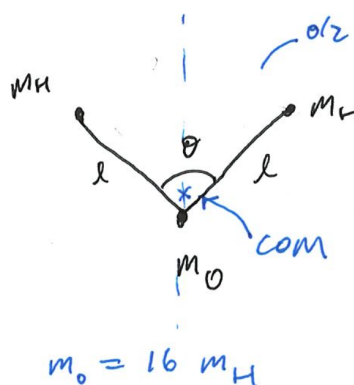
$$\begin{aligned}
 X &= \frac{A}{M} \left\{ \left[x \frac{2\ell}{\pi} \sin\left(\frac{\pi x}{2\ell}\right) \right]_0^\ell - \int_0^\ell \frac{2\ell}{\pi} \sin\left(\frac{\pi x}{2\ell}\right) dx \right\} \\
 &= \frac{A}{M} \frac{2\ell}{\pi} \left[\ell + \frac{2\ell}{\pi} \cos\left(\frac{\pi x}{2\ell}\right) \right]_0^\ell \\
 &= \frac{A}{M} \frac{2\ell}{\pi} \left(1 - \frac{2}{\pi} \right) \ell \\
 &= \ell \left(1 - \frac{2}{\pi} \right)
 \end{aligned} \tag{4.4}$$

The numeric factor evaluates to $X \approx 0.363\ell$. This matches the expectation that the mass is heavier on the left than the right, so the COM is pulled left of 0.5.

KK 4.1



KK 4.3



KK 4.3 - Water Molecule

We are given the dimensions of a H₂O molecule. Since it is left-right symmetric, the COM must lie on the line passing through the symmetry plane. It is convenient to define the Oxygen as the origin. Then, with

$\ell = 0.097$ nm and $\theta = 104.5^\circ$, we compute

$$X = \frac{\sum x_i m_i}{\sum m_i} = \frac{2x_H m_H}{2m_H + m_O} \quad (4.5)$$

Well $x_H = \ell \cos(\theta/2)$. We can take O-16 and H-1 as the most common isotopes of Oxygen and Hydrogen. Then $m_O = 16m_H$, and

$$X = \frac{2\ell \cos(\theta/2)}{2 + 16} = \frac{1}{9}\ell \cos(\theta/2) \quad (4.6)$$

Converting to radians, $\theta_{rad} = \frac{2\pi}{360}\theta_{deg} = 0.612$, and we find $X = 0.068\ell = 0.0066$ nm.

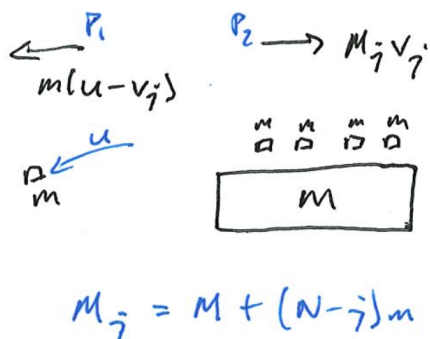
Let us note, although ℓ is given in nm = 10^{-9} m, it is more natural to express atomic lengths in the Angstrom $1\text{\AA} = 10^{-10}$ m = 0.1 nm. Then the covalent bond length is $0.97\text{\AA}$, and the COM is $0.066\text{\AA}$. We see that the COM is pulled very close to the Oxygen, as it must be.

Problem 5 - Extra Credit

KK 4.15 - Cart Jumpers

The key to the problem is conservation of momentum. N men of mass m each jump off a cart of mass M . In (a) they jump all at once, in (b) they jump one at a time. The central constraint is that the men jump with velocity u relative to the cart, not the ground.

KK 4.15



Note the signs: $u > v_j$

$$\begin{cases} P_1 = m(v_j - u) < 0 \\ P_2 = M_j v_j > 0 \end{cases}$$

a) simultaneous

Before the men jump the system is at rest. Therefore $P_i = 0$. After the men jump

$$P_1 = Nm(v - u) \quad (5.1)$$

$$P_2 = Mv \quad (5.2)$$

Momentum is conserved because there are no external forces. Then

$$P_f = P_1 + P_2 = Nm(v - u) + Mv = P_i = 0 \quad (5.3)$$

We wish to find the velocity of the cart v after the jump.

$$(Nm + M)v = Nmu \quad (5.4)$$

then

$$v = \left(\frac{Nm}{Nm + M} \right) u \quad (5.5)$$

Assuming $\epsilon = m/M \ll 1$, we recast this as

$$v = \left(\frac{N\epsilon}{1 + N\epsilon} \right) u \quad (5.6)$$

We see in the limit $N\epsilon \rightarrow 0$ that $v \ll u$. The cart moves slowly compared to the jumpers.

b) sequential

When the men jump one at a time, we get a sequence of masses

$$M_j = M + (N - j)m \quad (5.7)$$

after the j -th jump. Similarly, we may define v_j to be the velocity after the j -th jump. It follows that $v_0 = 0$, $M_N = M$, and we wish to find v_N .

Let's use conservation of momentum.

$$P_{j1} = m(v_j - u) \quad (5.8)$$

$$P_{j2} = M_j v_j \quad (5.9)$$

Then

$$P_f = P_{1j} + P_{2j} = m(v_j - u) + M_j v_j \quad (5.10)$$

but this time

$$P_i = M_{j-1} v_{j-1} \quad (5.11)$$

Then we find

$$(m + M_j)v_j = M_{j-1}v_{j-1} + mu \quad (5.12)$$

such that

$$v_j = \left(\frac{M_{j-1}}{m + M_j} \right) v_{j-1} + \left(\frac{m}{m + M_j} \right) u \quad (5.13)$$

can be solved recursively. Taking $v_0 = 0$, then

$$\begin{aligned} v_1 &= \left(\frac{m}{m + M_1} \right) u \\ v_2 &= \left(\frac{m}{m + M_1} \right) u + \left(\frac{m}{m + M_2} \right) u \\ v_3 &= \left(\frac{m}{m + M_1} \right) u + \left(\frac{m}{m + M_2} \right) u + \left(\frac{m}{m + M_3} \right) u \end{aligned} \quad (5.14)$$

et cetera. Recognizing the pattern, we conclude

$$v_N = \left[\left(\frac{m}{m + M_1} \right) + \left(\frac{m}{m + M_2} \right) + \cdots + \left(\frac{m}{m + M_N} \right) \right] u \quad (5.15)$$

c) comparison

We can write 5.5 as

$$\begin{aligned} v &= \left(\frac{Nm}{Nm + M} \right) u \\ &= \left[\left(\frac{m}{m + M_1} \right) + \left(\frac{m}{m + M_1} \right) + \cdots + \left(\frac{m}{m + M_1} \right) \right] u \end{aligned} \quad (5.16)$$

where $M_1 = M + (N - 1)m$. Since $M_0 > M_j$ for all $j > 1$, it follows that $v < v_N$. Jumping sequentially (b) always yields a greater velocity than jumping simultaneously (a).

For the edge case $N = 1$, we recover $v_1 = v$.

We can go further in analyzing 5.15. Expanding $M_j M + (N - j)m$, and substituting $\epsilon = m/M$

$$\begin{aligned} v_N &= \left[\left(\frac{m}{m + M} \right) + \left(\frac{m}{2m + M} \right) + \cdots + \left(\frac{m}{Nm + M} \right) \right] u \\ &= \left[\left(\frac{\epsilon}{1 + \epsilon} \right) + \left(\frac{\epsilon}{1 + 2\epsilon} \right) + \cdots + \left(\frac{\epsilon}{1 + N\epsilon} \right) \right] u \\ &= u \sum_{j=1}^N \frac{\epsilon}{1 + j\epsilon} \end{aligned} \quad (5.17)$$

In the limit where $\epsilon \rightarrow 0$, this sum becomes the integral

$$v_N = u \int_0^{N\epsilon} \frac{dx}{1 + x} = u \ln(1 + N\epsilon) \quad (5.18)$$

So the final velocity is a logarithmic function of N . This system approaches that of a rocket with a continuous discharge of particles. The relation is inverted to yield

$$N\epsilon = e^{v_N/u} - 1 \quad (5.19)$$

which says that increasing v_N/u requires exponentially more men.